

Department of Computer Science and Engineering
United International University
Assignment on Loops and Conditional Logics

Sr	Questions								
1	Dipto is working as a guard at Notunbazar XYZ gas Station. Cars arrive throughout the night, each needing a different amount of gas at a fixed price per liter. Your goal is to write a C program that calculates the total revenue earned at the end of the night, considering the following: • Number of cars: You don't know beforehand how many cars will arrive. Use a loop to handle an unknown number. • Gas amount: Each car needs a specific integer amount of gas (in liters). • Price per liter: The price per liter of gas is a fixed integer value. Challenge: Implement error handling for invalid input (example: negative liters, non-integer values). <table><tr><th>Sample Input</th><th>Sample Output</th></tr><tr><td>Enter the price per liter of gas: 150 Enter the number of cars: 3 Car 1 gas amount (litre): 20 Car 2 gas amount (litre): 35 Car 3 gas amount (litre): 40</td><td>Car 1: 20 liters * 150 = 3000 Car 2: 35 liters * 150 = 5250 Car 3: 40 liters * 150 = 6000 Total Revenue: 3000 + 5250 + 6000 = 14250</td></tr></table>	Sample Input	Sample Output	Enter the price per liter of gas: 150 Enter the number of cars: 3 Car 1 gas amount (litre): 20 Car 2 gas amount (litre): 35 Car 3 gas amount (litre): 40	Car 1: 20 liters * 150 = 3000 Car 2: 35 liters * 150 = 5250 Car 3: 40 liters * 150 = 6000 Total Revenue: 3000 + 5250 + 6000 = 14250				
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2	Expand the basic calculator to include floating-point numbers and operations like division with zero checks (conditional statements) and handling floating-point errors. Challenges: Checking for division by zero is crucial to avoid program crashes. Implementing proper error handling with informative messages is essential. <table><tr><th>Sample Input</th><th>Sample Output</th></tr><tr><td>2.5 + 3.14</td><td>5.64</td></tr><tr><td>10 / 0</td><td>Division by zero error!</td></tr><tr><td>-4.7 * 7.2</td><td>-33.84</td></tr></table>	Sample Input	Sample Output	2.5 + 3.14	5.64	10 / 0	Division by zero error!	-4.7 * 7.2	-33.84
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3	You're on a secret mission! You've intercepted a coded message, but it's only shifted a few positions forward in the alphabet (uppercase A-Z). Remember that you can not encode X,Y and Z. Challenge: Handle lowercase letters (a-z) with appropriate integer mapping. Shift value 3 means the ASCII value of the original message is 3 values less than the coded message. (For example, 'F' in coded message is actually 'C' in original).								

	<table><tr><th>Sample Input</th><th>Sample Output</th></tr><tr><td>Message: FRPH PHHW ZLWK DJHFW MDQQDWXO Shift value: 3</td><td>COME MEET WITH AGENT JANNATUL</td></tr></table>	Sample Input	Sample Output	Message: FRPH PHHW ZLWK DJHFW MDQQDWXO Shift value: 3	COME MEET WITH AGENT JANNATUL		
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4	You're building a program to identify all Digit-Primes within a specific range. How would you approach this task? Challenges: Take input from user If the sum of individual digits of a number is a prime number then the number itself is a “Digit Prime”. <table><tr><th>Sample Input</th><th>Sample Output</th></tr><tr><td>Enter two numbers(intervals): 100 120</td><td>Digit Prime between 100 and 120 are: 100 101 102 104 106 110 111 113 115 119 120</td></tr></table>	Sample Input	Sample Output	Enter two numbers(intervals): 100 120	Digit Prime between 100 and 120 are: 100 101 102 104 106 110 111 113 115 119 120		
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5	Write a program in C that checks if an input integer is a perfect number or not. A perfect number is defined as an integer for which the sum of all its positive divisors (excluding itself) equals the number itself. For example, 6 is a perfect number because 1+2+3=6. Your program should read integer numbers and print a message indicating whether they are perfect or not. <table><tr><th>Sample Input</th><th>Sample Output</th></tr><tr><td>6</td><td>6 is a perfect number</td></tr><tr><td>10</td><td>10 is not a perfect number</td></tr></table>	Sample Input	Sample Output	6	6 is a perfect number	10	10 is not a perfect number
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6

Write a program that converts Celsius into Fahrenheit temperatures.

Formula: $F = \{ C * (9/5) \} + 32$

Challenges:

Handle negative temperatures and temperatures outside the normal range.

Sample Input	Sample Output
0 °C	32 °F
20 °C	68 °F
80 °C	176.0 °F

Note: No need to show °C and °F unit. Only calculation

$$^{\circ}F = ^{\circ}C \times 9/5 + 32$$